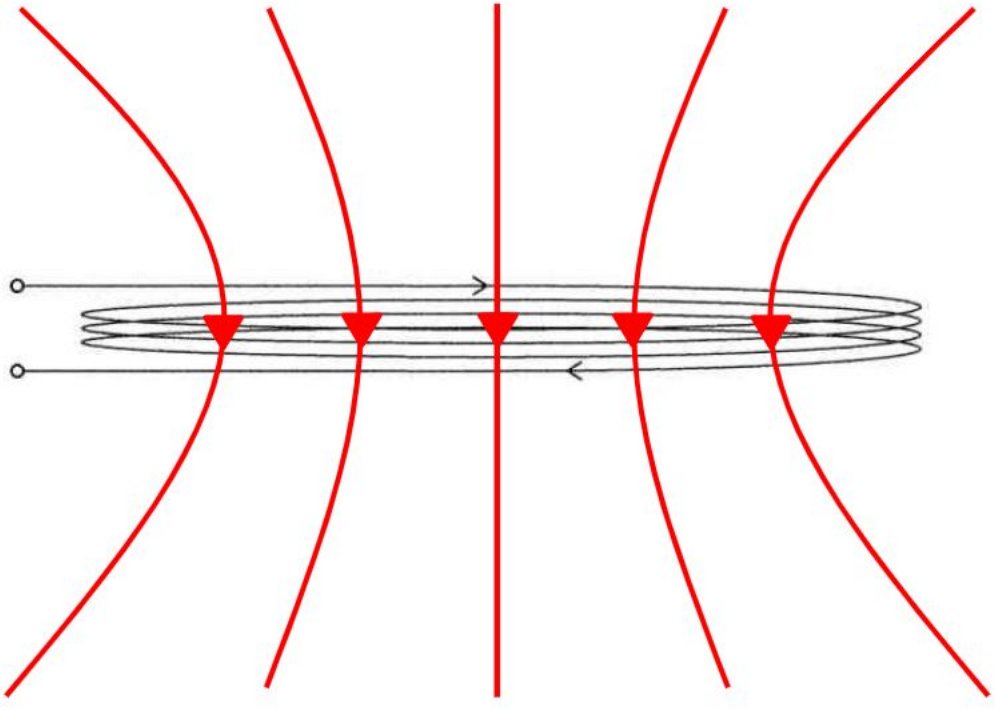
8 **(a)** Magnetic flux density is the force acting per unit current per unit length on a conductor placed perpendicular to a magnetic field.

(b) (i)



*correct direction, symmetrical about the centre, at least 5 lines occupying the entire coil - straight line in the centre, other 4 lines to spread outwards, different radius. The further the line from the centre, the smaller the radius of the arc

(ii) **1.** The current in the wire is pointing to the left and the magnetic flux density is acting into the page. By Flemming's Left Hand Rule, the wire experiences a magnetic force vertically downwards perpendicular to both the current and the magnetic field when viewed from the top.

Hence, the wire moves downwards.

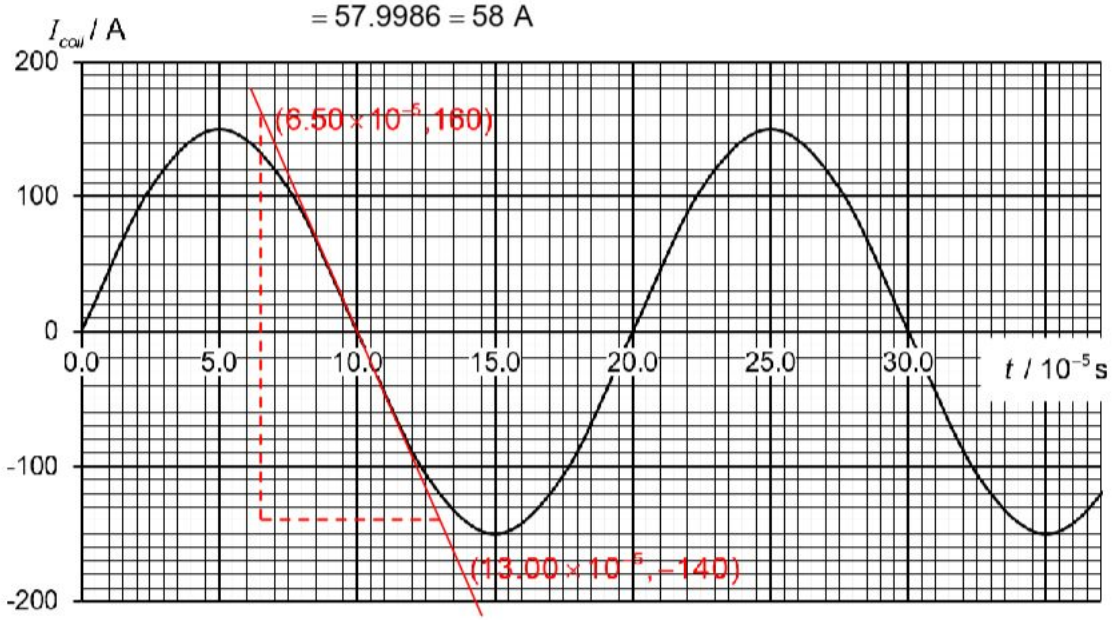
2. There is no effect on the wire as the direction of the magnetic field of the coil along the axis is parallel to the direction of the current in the wire.

(c) (i) The high frequency alternating current sets up a high frequency oscillating magnetic field in the flat circular coil. As a result, the magnetic flux through the base of the saucepan changes at a high frequency.

According to Faraday's law, a large emf is induced in the base of the saucepan. Since the base of the pan is a conductor with resistance, a large eddy current is produced. This gives rise to Joule heating and a large amount of thermal energy is dissipated, heating the contents in the saucepan.

(ii) **1.**

$$\begin{aligned}(I_{induced})_{max} &= \frac{|E_{max}|}{R} \\ &= \frac{1}{R}\left|-\frac{d\phi}{dt}\right|_{max} \\ &= \frac{1}{R}\left(\frac{dBA}{dt}\right)_{max} \\ &= \frac{A}{R}\left(\frac{dB}{dt}\right)_{max} \\ &= \frac{A}{R}\left(\frac{d(5\mu_o N I_{coil})}{dt}\right)_{max} \\ &= \frac{5\mu_o NA}{R}\left(\frac{dI_{coil}}{dt}\right)_{max} \\ &= \frac{5(4\pi\times10^{-7})(20)(0.020)}{(0.20)}\left|\left(\frac{-140-160}{13.00\times10^{-5}-6.50\times10^{-5}}\right)\right| \\ &= 57.9986 = 58 \text{ A}\end{aligned}$$

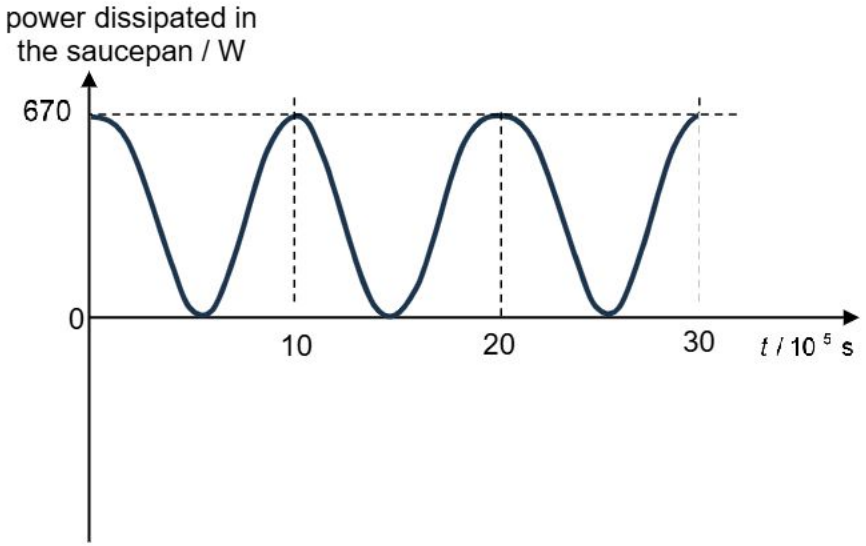


OR

$$\begin{aligned}(I_{induced})_{max} &= \frac{|E_{max}|}{R} \\ &= \frac{1}{R}|B_{max}A\omega| \\ &= \frac{1}{R}(5\mu_o N I_{coil,max})(A)\left(\frac{2\pi}{T}\right) \\ &= \frac{(5)(4\pi\times10^{-7})(20)(150)(0.020)(2\pi)}{(0.20)(20\times10^{-5})} \\ &= 59.23 = 59 \text{ A}\end{aligned}$$

2.

$$\begin{aligned}P_o &= (I_{induced,max}{}^2 R) \\ &= (57.9986^2)(0.20) \\ &= 672.77 = 670 \text{ W}\end{aligned}$$



*cosine squared graph with correct period and value of peak power.

3.

$$\langle P \rangle = \frac{1}{2} P_o$$

$$Q = \langle P \rangle t = mc\Delta T$$

$$\Delta T = \frac{\langle P \rangle t}{mc}$$

$$= \frac{0.50 \times 672.77 \times (40 \times 60)}{0.900 \times 2000}$$

$$= 44.851 = 45 \ ^\circ\text{C}$$

(iii) The high magnetic permeability of stainless steel maximises the (rate of change of) magnetic flux linkage between the flat circular coil and base of the cookware. This results in large induced emf / eddy currents induced in the cookware.